BHAVYA KOHLI

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Education

National University of Singapore — Singapore
PhD in Electrical and Computer Engineering — *Aug'25 – Present*

Indian Institute of Technology (IIT) Bombay — Mumbai, India
B.Tech in Electrical Engineering and M.Tech in Artificial Intelligence ***(Cumulative GPA: 9.13/10.0)*** — *Nov'20 – Aug'25*

- ***Key Coursework:*** *Machine Learning, Optimization, Graph Models, Graph Learning, Probability, Linear Algebra*

Publications

1. **Bhavya Kohli**, Aziz Shameem, Abir De "**Learning Right Monotone Permutation Matrices for Neural Subsequence Search**", *accepted* at The Twenty-Ninth Annual Conference on Artificial Intelligence and Statistics (AISTATS 2026)
2. Prateek Garg*, **Bhavya Kohli***, Sunita Sarawagi "**Masked Diffusion Models are secretly Learned-Order Autoregressive Models**", *published* at the EurIPS 2025 Workshop on Principles of Generative Modeling (PriGM). Preprint available at arXiv:2511.19152
3. **Bhavya Kohli**, Varun Kohli, Muhammad Naveed Aman, Biplab Sikdar "**Swarm-Net: Firmware Attestation in IoT Swarms using Graph Neural Networks and Volatile Memory**", *published* at the IEEE Internet of Things Journal. Preprint available at arXiv:2408.05680
4. Julia Werner, **Bhavya Kohli**, Paul Palomero Bernardo, Christoph Gerum, Oliver Bringmann "**Energy-Efficient Seizure Detection Suitable for Low-Power Applications**", *published* at the 2024 International Joint Conference on Neural Networks (IJCNN), IEEE, 2024. Preprint available at arXiv:2406.16948

Research Experience

Graph Summarization for Tackling Graph-Level Tasks — *Aug'24 – Aug'25*
Master's Thesis, Guide: Prof. Abir De — *CSE, IIT Bombay*

- **Introduction:** *Many algorithms for difficult graph-level tasks scale poorly with the size of the input graph(s). The aim of the project is to devise algorithms to compress graphs to aid in both efficient execution, and training new models on these tasks.*
- Organized existing literature on graph summarization with the goal of inducting useful components based on the setting
- Implemented a set of baseline algorithms from literature and applied them to specific tasks to gain deeper insights into the problem

Firmware Attestation in IoT Swarms using Volatile Memory — *May'24 – Aug'24*
Research Project, Guide: Prof. Dr. Biplab Sikdar — *ECE, National University of Singapore*

- **Introduction:** *Malicious activity on one node in a swarm can propagate to larger network sections, possibly impacting other clusters of devices. The proposed method works by leveraging the IoT devices' Static random-access memory (SRAM) making it lightweight, efficient, and unaffected by the usual drawbacks of attestation methods.*
- Conducted an extensive survey on modeling **IoT swarms** to best utilize graph models in the restricted-information setup
- Designed and implemented a **graph transformer network** among other popular baseline graph convolution operators
- Evaluated **different adversarial scenarios** inspired from real-world applications and commonly encountered edge cases

Sequence Retrieval using Regularized Soft Permutation Matrices — *Aug'23 – Aug'24*
Bachelor's Thesis, Guide: Prof. Abir De — *CSE, IIT Bombay*

- **Introduction:** *Information retrieval often involves complex and intensive computations for getting a ranked list of corpus elements with decreasing relevance to a given query. The proposed plug-and-play method uses permutation matrices as a basis for query-corpus alignment which is followed by simple metric computations.* ***The manuscript for this work is in progress.***
- Modeled an optimization problem to learn unknown permutations using the **linear sum assignment** problem as a basis
- Derived a concise representation of the objective which allowed for appropriate neural replacements of key optimization steps
- Implemented workflows for testing the end-to-end system using multiple sequence datasets with different embedding strategies

Online Hardware-aware Seizure Detection — *May'23 – Aug'23*
Research Project, Guide: Prof. Dr. Oliver Bringmann — *CSE (Embedded Systems), University of Tübingen, Germany*

- **Introduction:** *Hardware-aware machine learning allows for seamless deployment of algorithms on low-power edge devices. The proposed seizure detection approach involving a low-footprint TC-ResNet combined with time-series analysis is extremely energy efficient and is suitable for real-time seizure detection using neural implants and wearable devices.*
- Performed a thorough literature survey on classical time-series processing methods and classification approaches
- Contributed to an **open-source hardware acceleration library** (**HANNAH**) for hardware-aware machine learning
- Implemented a time-series-specific data processing pipeline, leading to a $> 92\%$ **sensitivity** on unseen seizures

Security of Graph Neural Networks in IoT — *May'22 – Jul'22*
Research Project, Guide: Prof. Dr. Biplab Sikdar — *ECE, National University of Singapore*

- **Introduction:** *Machine Learning models trained without explicit defense mechanisms are susceptible to adversarial attacks. We explore graphical models, their benefits over others in the IoT domain, which often deals with networks of interconnected sensors, and their robustness against black-box adversarial attacks.*
- Studied and evaluated black-box adversarial attacks and defenses on convolutional neural networks and graph neural networks
- Surveyed and implemented query-efficient **black-box gradient estimation** based adversarial attack on Spatio-Temporal GCNs
- Performed an in-depth attack efficacy comparison with auxiliary model attacks and a few other black-box methods from literature

Scholastic Achievements

- Accepted into the IDDD Programme at the **Center for Machine Intelligence and Data Science** *(2023)*
- Secured an **All India Rank of 675** in JEE Advanced (Among top 0.33% in the country) *(2020)*
- Part of the **99.848 percentile** in the JEE Mains 2020 (Among top 0.3% in the country) *(2020)*
- Awarded the **Kishor Vaigyanic Protsahan Yojana (KVPY)** fellowship (SX Stream) *(2020)*
- Scored **383/450** in the Birla Institute of Technology and Science Admission Test (BITSAT) *(2020)*

Key Technical Projects

Local Augmentation for Graph Neural Networks *Aug'23 – Nov'23*
Course Project, Guide: Prof. Abir De (CS768: Learning with Graphs) *IIT Bombay*

- Implemented the local augmentation method described in the assigned paper, and extended the method to other graph tasks—**edge prediction and graph-level classification**. Observed an improvement over non-augmented baselines
- Implemented a **normalizing flow** model demonstrating superior generative capability at the cost of compute-efficiency
- Studied the over-smoothing effect in GNNs and experimented with **probabilistic message-passing** to combat it

Representation Learning using Spiking Autoencoders *Sep'23 – Nov'23*
Course Project, Guide: Prof. Udayan Ganguly (EE746:Neuromorphic Engineering) *IIT Bombay*

- Studied spike and rate and spike coding techniques used in **neuromorphic neural models** for tasks involving structured data
- Studied the possible benefits of neuromorphic models over existing paradigms in energy efficiency and robustness
- Implemented a **spiking neural network** (SNN) based autoencoder as a generative model for a few baseline datasets

Variational Thompson Sampling *Jan'23 – May'23*
Course Project, Guide: Prof. Jayakrishnan Nair (EE6106: Online Learning and Optimisation) *IIT Bombay*

- Explored **variational inference** for approximating posteriors in the **Multi-Armed Contextual Bandit** problem
- Studied a **Gaussian mixture model** (GMM) to model the distribution of parameters over arms
- Studied trade-offs between the variational approach and using Gibbs sampling for this model

Unsupervised Representation Learning using Tensorized Autoencoders *Nov'22 – Dec'22*
Self project

- Studied methods for unsupervised representation learning using autoencoders and other generative models (VAEs)
- Implemented a new “tensorized autoencoder" architecture regularized using clustering algorithms and benchmarked it using a standard AE on clustering, data denoising, and vision tasks such as image inpainting and denoising

A study on the Learning with Errors problem *Oct'22 – Dec'22*
Course Project, Guide: Prof. Bernard Menezes (CS741: Advanced Network Security and Cryptography) *IIT Bombay*

- Studied the Learning with Errors problem and its versatility of use in designing cryptographic schemes
- Implemented a few attacks on binary LWE and a public key based cryptographic scheme using text as the test medium

Technical Skills

Programming Languages	Python, Julia, C/C++
Machine Learning Libraries	PyTorch, Pytorch-geometric, Tensorflow, scikit-learn, Keras, snnTorch
Misc. Software and Tools	LaTeX, Git, Secure Shell (SSH), MATLAB, Docker, Bash scripting

Teaching Experience and Mentorship

Student Mentor
Institute Student Mentorship Program: Selected through rigorous interviews and peer reviews *2023-2024*

- Responsible for guiding 10 freshmen through their first year at IITB as an **Institute Student Mentor**
- Mentored a group of 5 undergradutes in their sophomore year with their academics as a **Department Academic Mentor**

Graduate Teaching Assistant
*TA for the course **CS419: Introduction to Machine Learning** for Prof. Abir De, CSE* *Spring'24*

- Mentored a class of **100+** cross-department students as part of a 7-member team
- Assisted the instructor by setting assignments, mentoring students, and conducting tutorials and help sessions

Extracurricular Activities

Achievements	• Certified for the Management Bootcamp and LaTeX courses conducted under the Non-Technical Summer School (NTSS) initiative by UGAC, IIT Bombay in 2021 • Won 2nd place as one of eight contingents in the Inter IIT Scrabble League (IISL) conducted in July 2021
Volunteering	• Devoted **80+ hours of volunteering work** and awarded a special mention for Exemplary Volunteering Work with under the National Service Scheme, IIT Bombay for the duration Dec'20 to Jul'21 • Volunteered to teach python to over 500 undergraduates as part of a team over the summer in 2023
Sports	• Completed a trek spanning **6+ hrs** to Anjaneri Hill Fort, Nashik, Maharashtra **4263 ft** above the sea level